Marlow Eliot

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Full UK right to work (British and Italian citizen)

MSc student in **AI for Sustainable Development** at UCL, with a **First** in Computer Science and AI. Interested in **applied machine learning and research**, building systems that find structure in messy real-world data, as well as **AI governance, safety and policy**.

EDUCATION

University College London | *MSc Artificial Intelligence for Sustainable Development* Sep 2026 – Sep 2027 (expected)

- Modules including: **Accountable, Transparent and Responsible AI**; Deep Representations and Learning; Probabilistic Modelling; Applied Artificial Intelligence.

University of Sussex | *BSc (Hons) Computer Science and Artificial Intelligence*, ***First-Class Honours*** Sep 2023 – Jun 2026

- Dissertation: **88/100** (see Projects).
- Selected to represent the university in a Collaborative Online International Learning (COIL) debate on the societal impacts of AI.

PROJECTS

Self-Supervised Behavioural Analysis Pipeline (BSc Dissertation) | *Python, PyTorch, HDBSCAN, UMAP* 2025 – 2026

- Built an end-to-end pipeline that discovers behavioural states in **7.4 million frames** of 3D movement data from **52** tracked bumblebees, with **no manual labelling**.
- Learned movement representations with a convolutional autoencoder and clustered them into behaviours with UMAP and HDBSCAN.
- Ran a systematic architecture search that cut reconstruction loss to **~1.5% of baseline**; using silhouette score, seed stability and noise %, showed **better reconstruction does not guarantee more stable clusters**.

Evolving Flocking Behaviour | *Python, PyBullet, evolutionary algorithms, NumPy* 2025

- Rebuilt Reynolds' boids model in a PyBullet physics simulation, with randomly placed obstacles to prevent overfitting.
- Benchmarked **four** evolutionary algorithms (hill climber, microbial GA, spatial GA, island-model GA) on tuning flock parameters.
- Proposed a new stratified model with separate rules for edge and inner agents, which **outperformed the standard model under all four algorithms**.

Environmental Degradation Database | *Assisted a World Vision Technical Advisor* Jan 2026 – Apr 2026

- Built a spreadsheet database for **AI agents**, linking each environmental issue to evidence from tracking data, surveys, studies and papers, with hidden instruction rows telling agents exactly what each column should contain.
- Piloted it on **Armenia** with the advisor for the Middle East and West Asia region, designed to help field teams identify local issues and choose responses.

EXPERIENCE

The Badger, University of Sussex student newspaper | *Senior Print Editor and Treasurer* Jul 2025 – May 2026

- Led editing and production for a monthly paper with **150+ writers** and a **1,500-copy print run**.
- Managed a **~£4,000 budget** as Treasurer, and helped run fundraisers, including the Save the Badger music event, to keep the paper running.

The Badger, University of Sussex student newspaper | *Science and Technology Editor* Jul 2024 – May 2025

- Led the section to a **nomination for Best Science and Technology Publication** at the 2026 SPA Awards.

RSPCA, Huddersfield branch | *Web Design Work Experience* Jun 2022

- Reviewed the branch website and proposed a redesign of its adoption pages.

SELECTED WRITING

- “Does One ChatGPT Prompt Really Use a Bottle of Water?”, *The Badger*, May 2026.
- “Running out of Data?”, *The Badger*, Sep 2024.
- “The TikTok Ban, ByteDance, and the ‘Splinternet’” (with Harry Turnbull), *The Badger*, Feb 2025.
- Co-founder, The Pasturage Institute (*Substack*): non-partisan essays on politics, philosophy, and policy, 2026.

SKILLS

- **Machine learning:** PyTorch, scikit-learn, self-supervised/unsupervised learning, clustering (HDBSCAN, UMAP), evolutionary algorithms.
- **Programming:** Python, NumPy, pandas, SQL, JavaScript, HTML/CSS, Git, PyBullet.
- **Other:** Writing and editing, budgeting, public debate, bouldering.